Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	08 March 2026
Team ID	699d5f9ac40b5ad0e9ee5a30
Project Name	QUOTES RECOMMENDATION CHATBOT USING NLP
Maximum Marks	4 Marks

Technical Architecture:

The system architecture of the Quotes Recommendation Chatbot consists of a Streamlit-based web interface that allows users to interact with the chatbot. The user messages are sent to the backend through a REST API. The Rasa NLP engine processes the message, classifies the user intent using the trained machine learning model, and selects the appropriate quote response. The system also supports features like saving favorite quotes using an SQLite database. The chatbot then returns the response to the frontend interface, where it is displayed to the user.

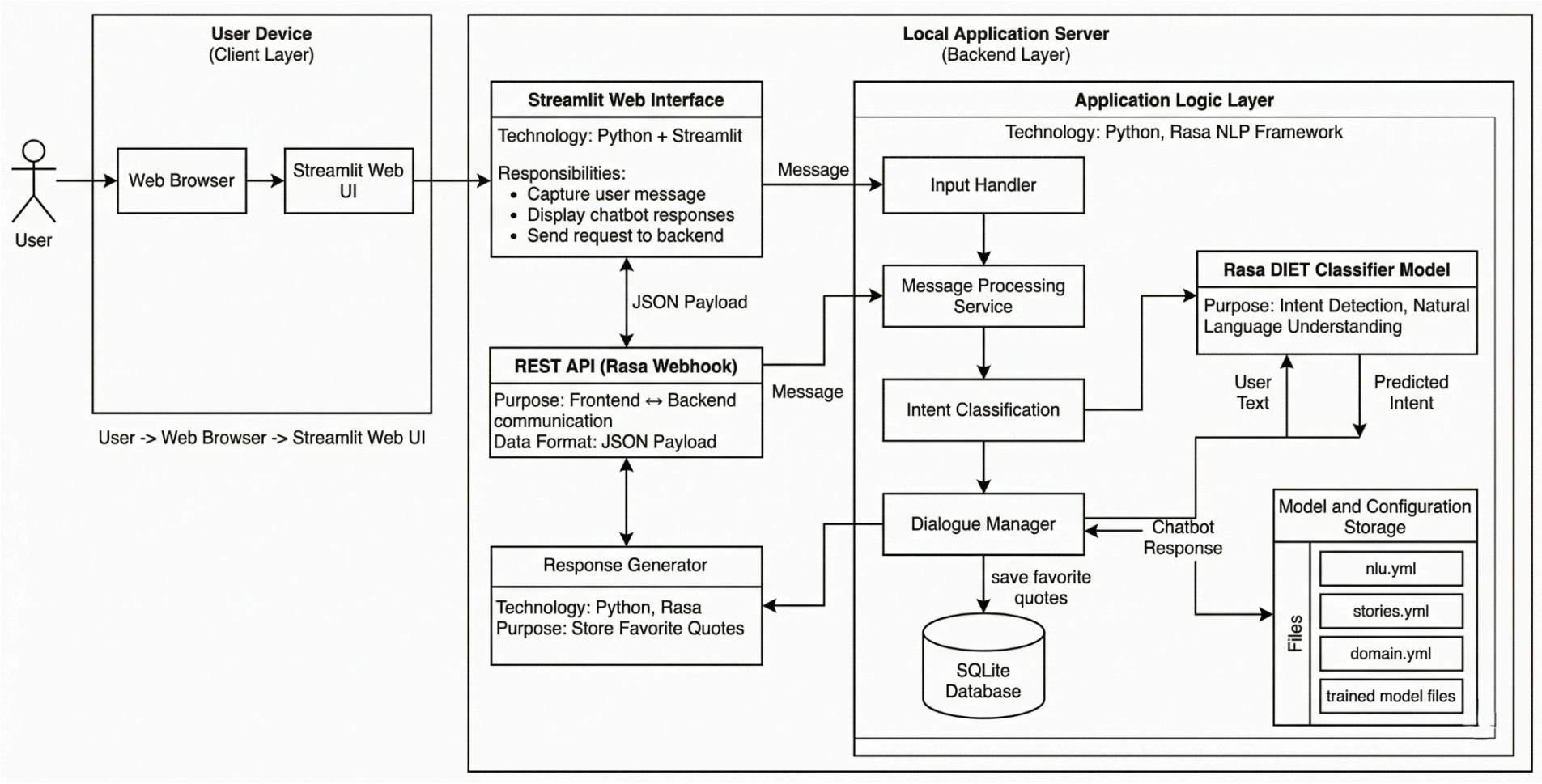
User Device
(Client Layer)
User
Web Browser
Streamlit Web UI
User -> Web Browser -> Streamlit Web UI
Local Application Server
(Backend Layer)
Streamlit Web Interface
Technology: Python + Streamlit
Responsibilities:
• Capture user message
• Display chatbot responses
• Send request to backend
Message
JSON Payload
REST API (Rasa Webhook)
Purpose: Frontend ↔ Backend communication
Data Format: JSON Payload
Message
Response Generator
Technology: Python, Rasa
Purpose: Store Favorite Quotes
Application Logic Layer
Technology: Python, Rasa NLP Framework
Input Handler
Message Processing Service
Intent Classification
Dialogue Manager
save favorite quotes
SQLite Database
Rasa DIET Classifier Model
Purpose: Intent Detection, Natural Language Understanding
User Text
Predicted Intent
Chatbot Response
Model and Configuration Storage
Files
nlu.yml
stories.yml
domain.yml
trained model files

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface where users interact with chatbot	Streamlit
2.	Application Logic-1	Handles user input and frontend interaction	Python
3.	Application Logic-2	NLP processing and intent classification	Rasa NLP
4.	Application Logic-3	Dialogue management and response generation	Rasa Core
5.	Database	Stores favorite quotes and session data	SQLite
6.	Cloud Database	Not used (local database used)	SQLite (Local)
7.	File Storage	Stores chatbot training data and models	Local File System
8.	External API	REST API for communication between frontend and backend	Rasa REST API
9.	Machine Learning Model	Intent classification and dialogue prediction	DIET Classifier (Rasa)
10.	Infrastructure (Server / Cloud)	Application deployment environment	Local Server / Python Environment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for chatbot development	Rasa, Streamlit
2.	Security Implementations	Secure communication between frontend and backend	REST API, Input validation
3.	Scalable Architecture	System designed with modular architecture	NLP backend + Web frontend
4.	Availability	Chatbot accessible through web browser anytime	Local server deployment
5.	Performance	Fast response time using pre-trained NLP models	Rasa NLP Engine